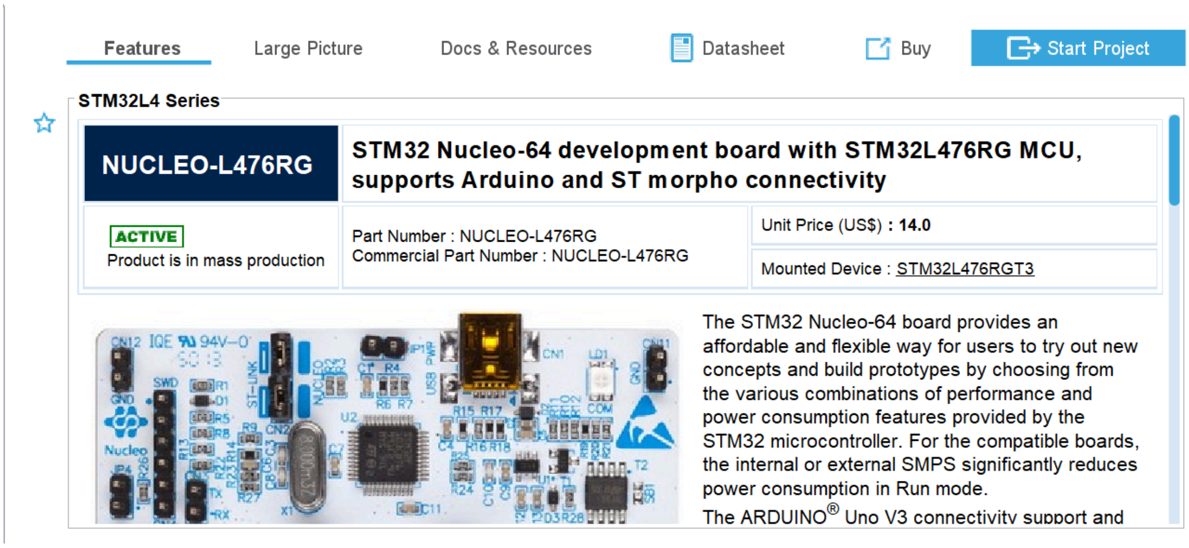
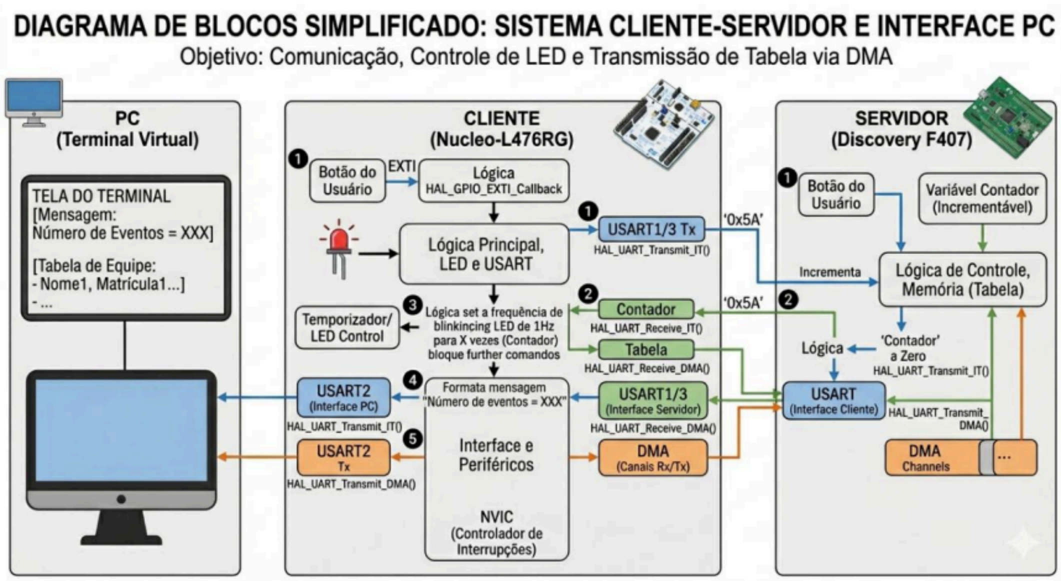
Configuração dos Projetos

Projetos do STM32CUBEmx



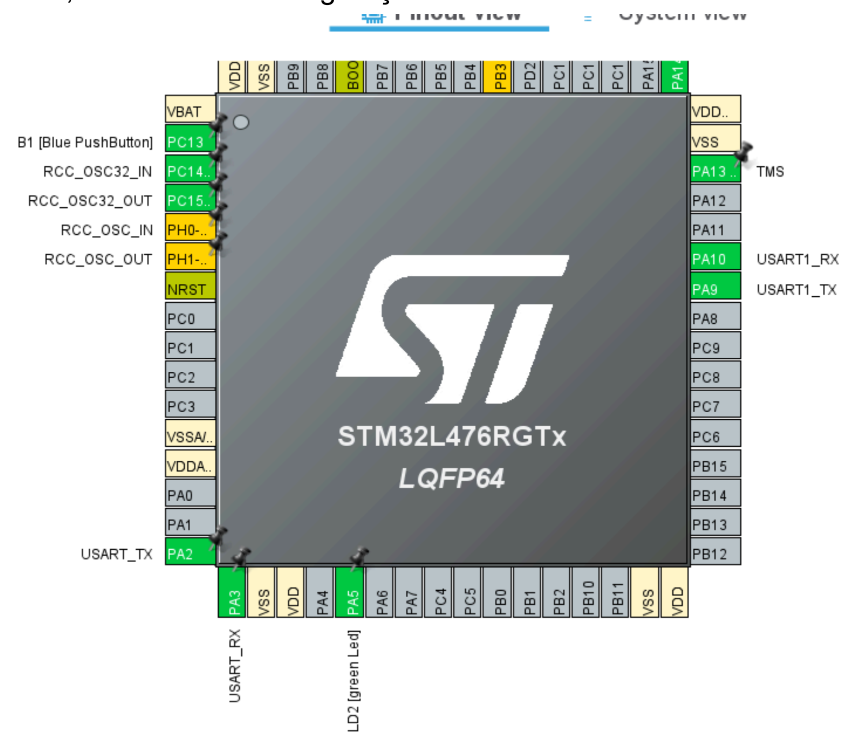
Configuração NUCLEO (L476RG)

O Projeto do professor solicita:

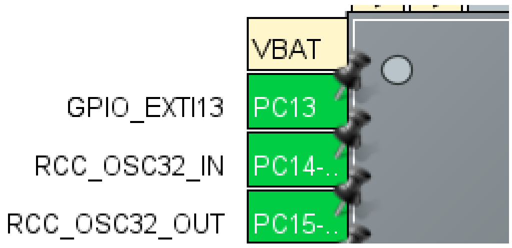


(Imagem gerada por IA)

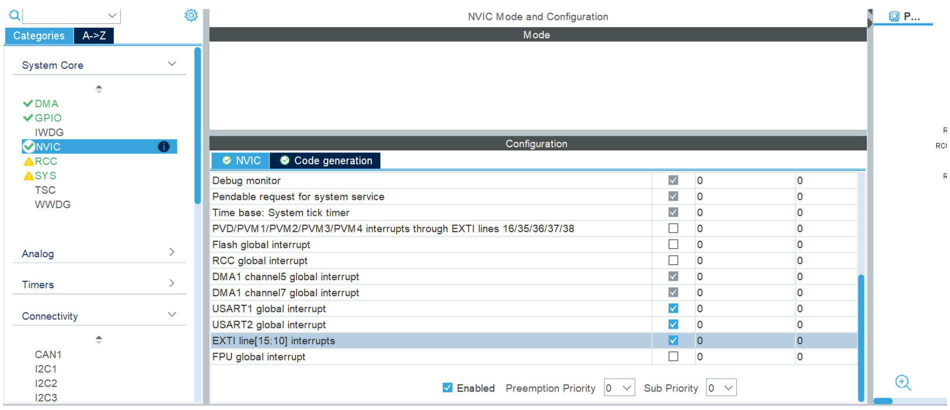
por isso, adotamos as configurações:



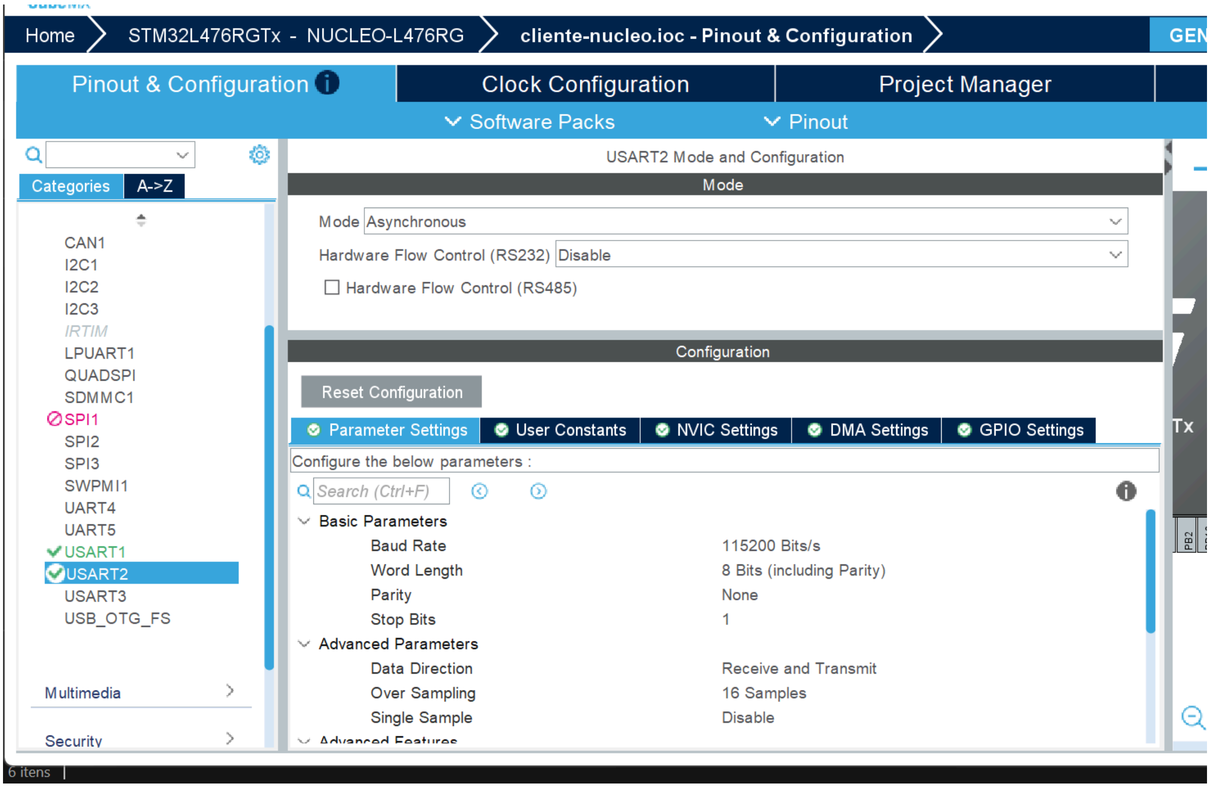
realizamos a substituição do pin PC13

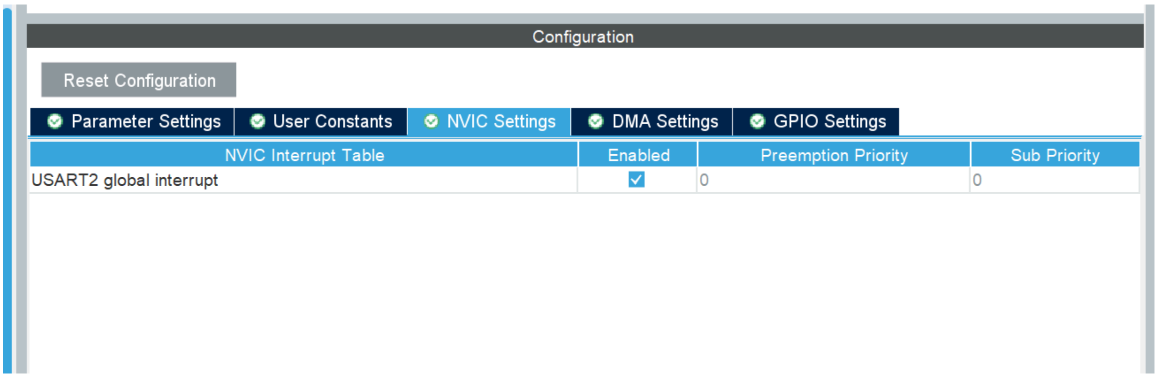


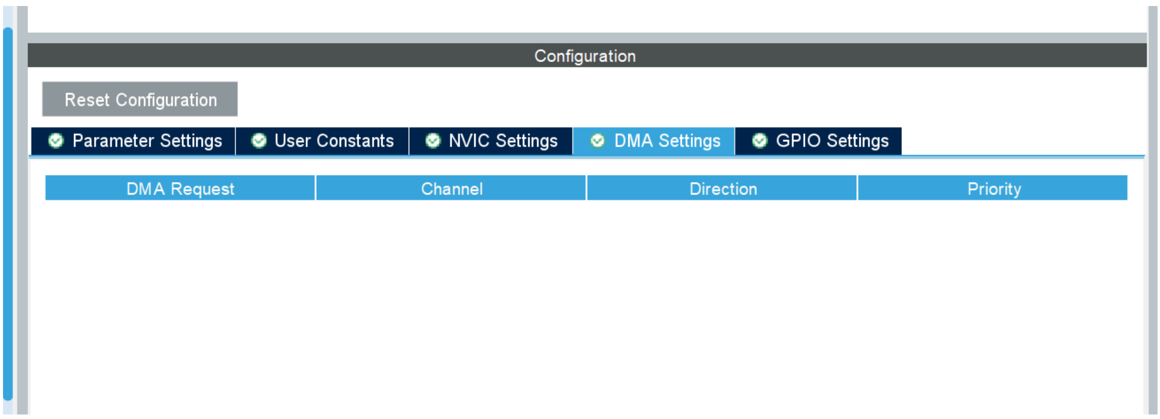
por fim, marquei o NVIC

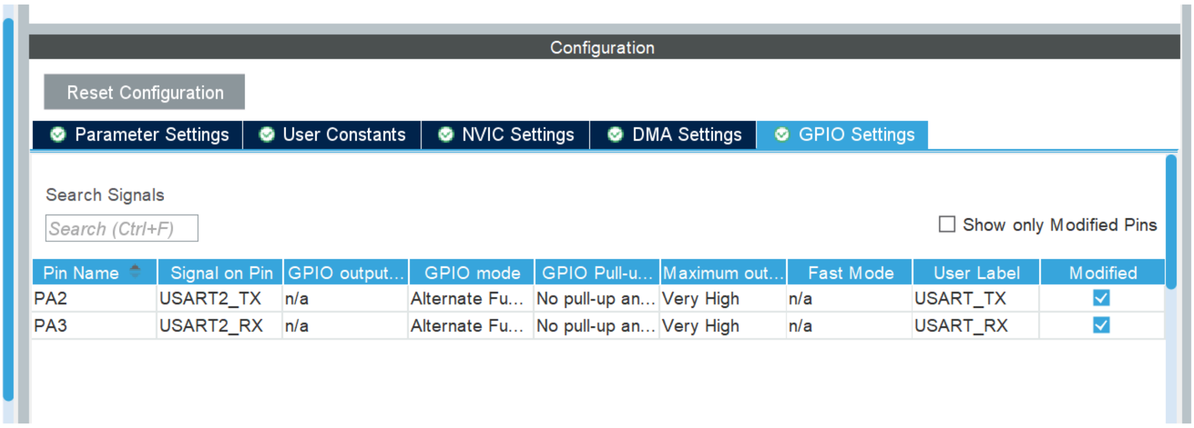


Para USART2, que será ligada com a maquina:

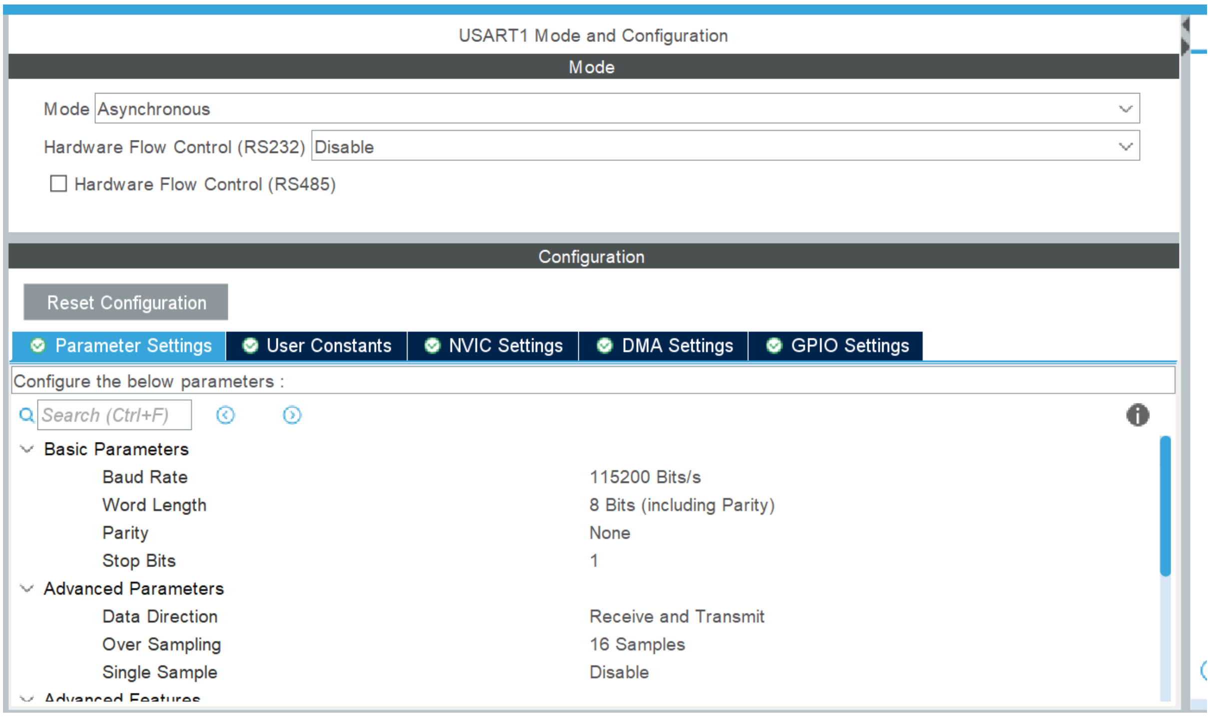


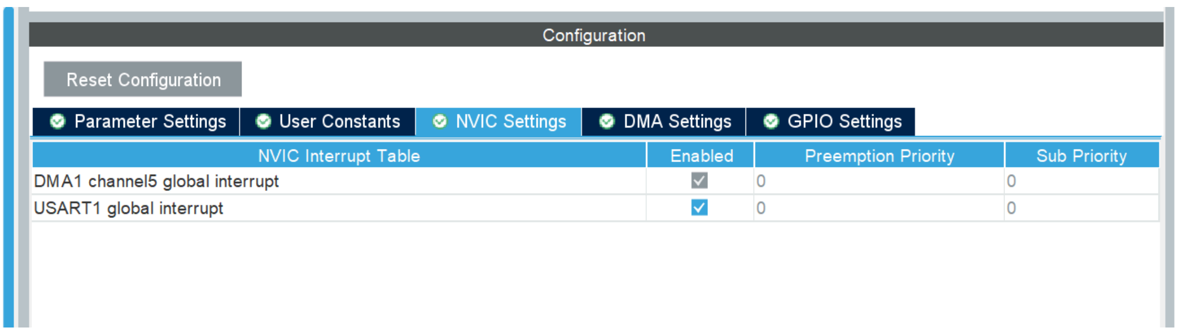


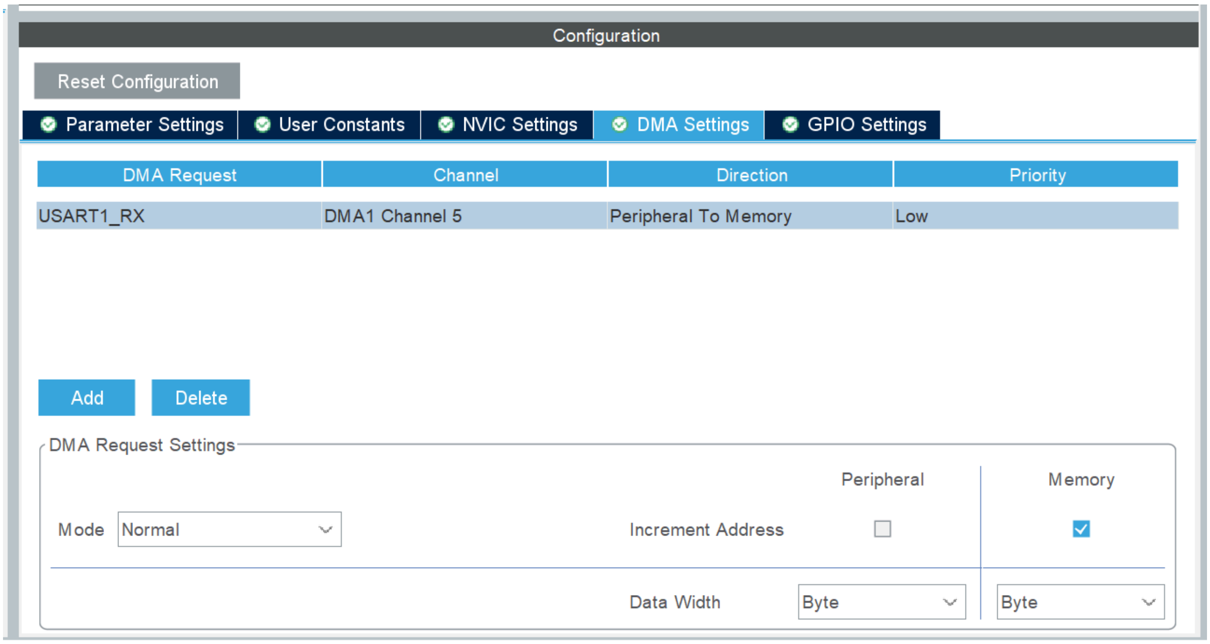


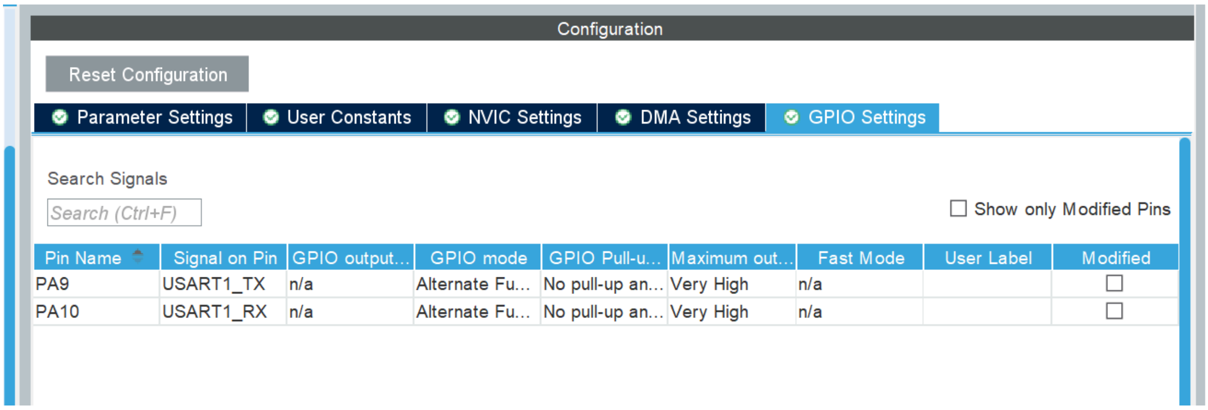


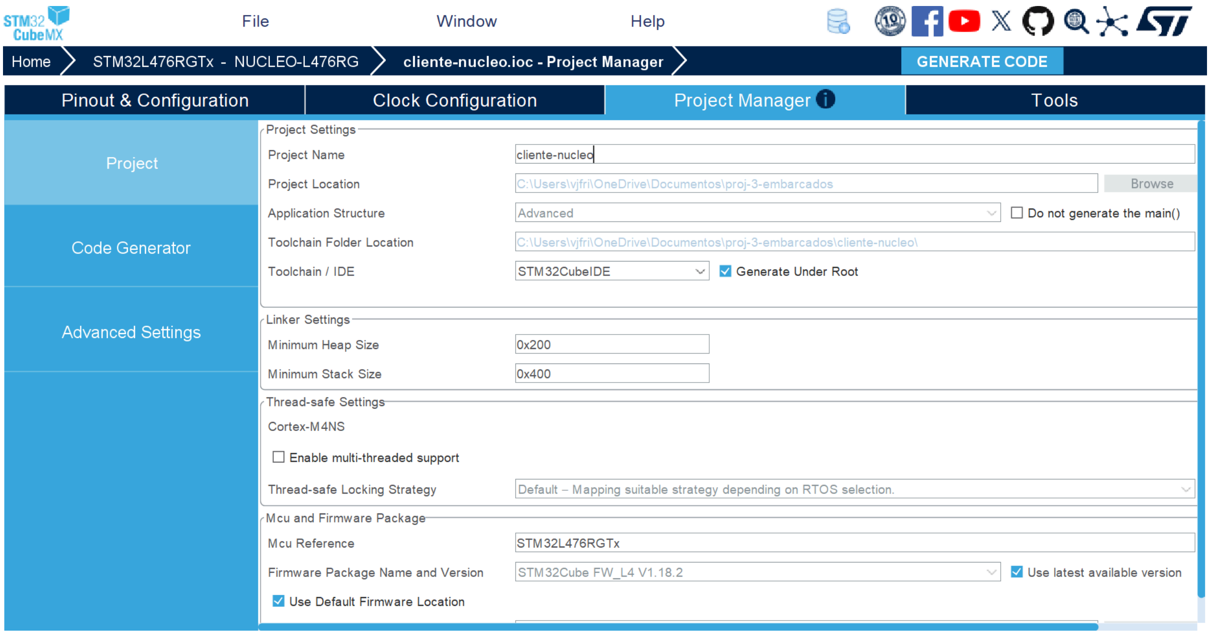
Para USART1, que será ligado na Discovery:





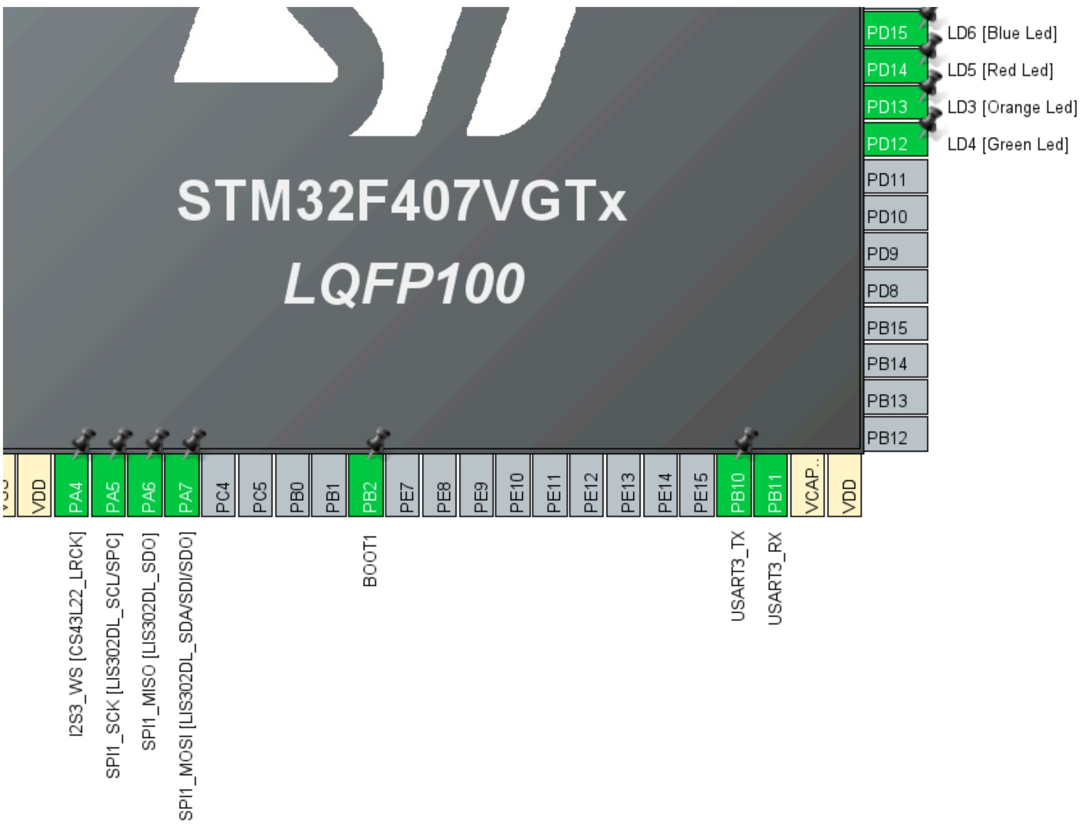

Configuration
Reset Configuration
Parameter Settings
User Constants
NVIC Settings
DMA Settings
GPIO Settings
DMA Request
Channel
Direction
Priority
USART1_RX
DMA1 Channel 5
Peripheral To Memory
Low
Add
Delete
DMA Request Settings
Peripheral
Memory
Mode
Normal
Increment Address
Data Width
Byte
Byte


Configuration
Reset Configuration
Parameter Settings
User Constants
NVIC Settings
DMA Settings
GPIO Settings
Search Signals
Search (Ctrl+F)
Show only Modified Pins
Pin Name
Signal on Pin
GPIO output...
GPIO mode
GPIO Pull-u...
Maximum out...
Fast Mode
User Label
Modified
PA9
USART1_TX
n/a
Alternate Fu...
No pull-up an...
Very High
n/a
PA10
USART1_RX
n/a
Alternate Fu...
No pull-up an...
Very High
n/a

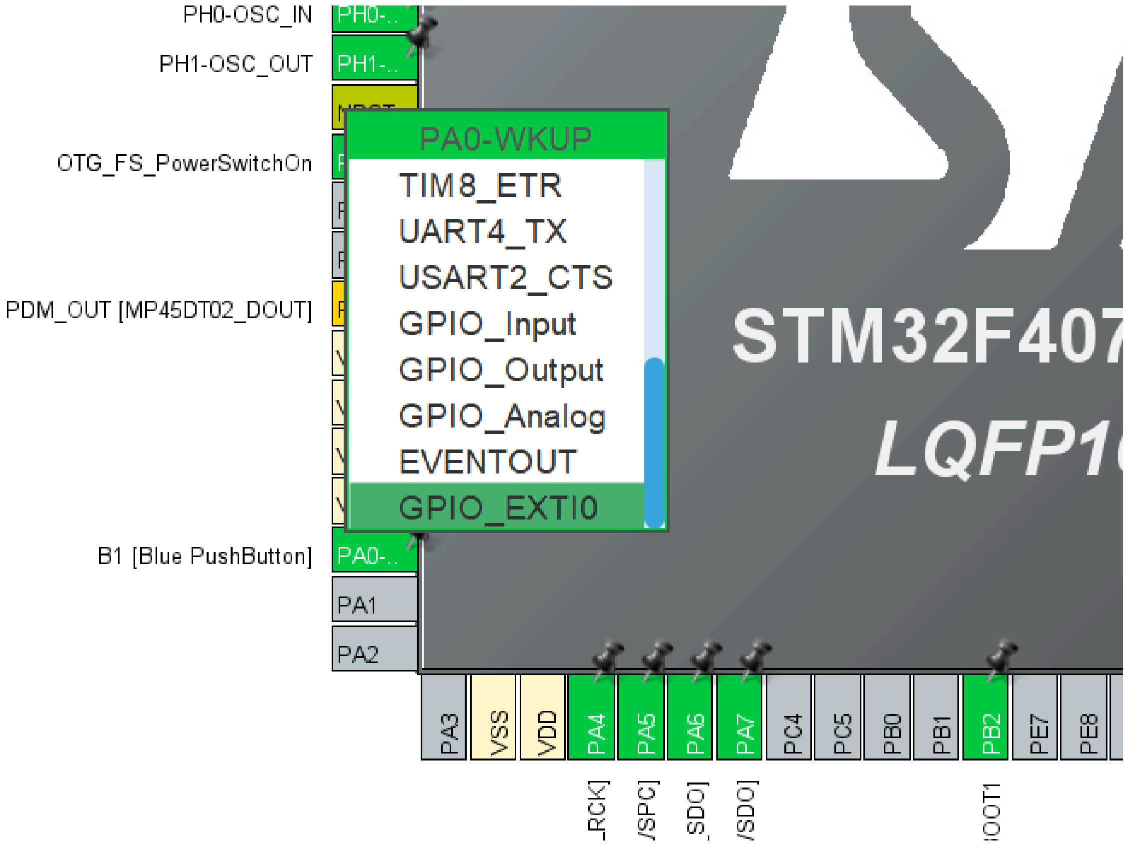


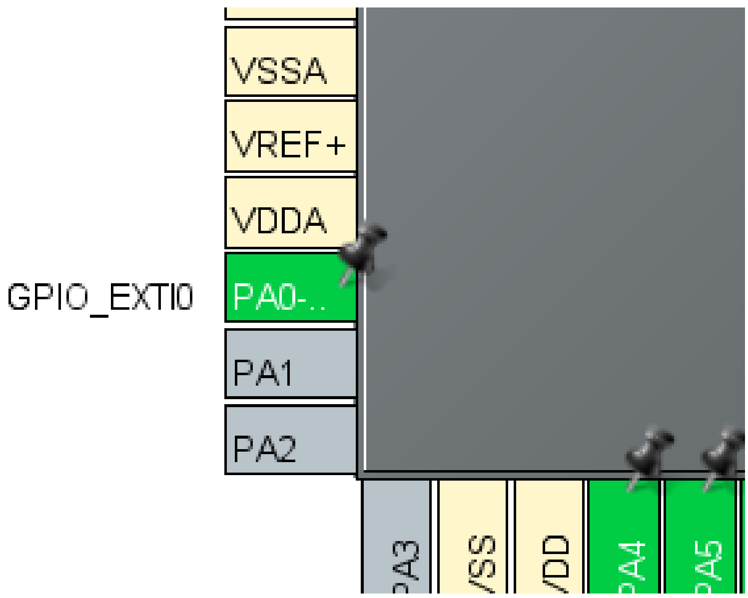
(OBS: sempre gerar código e colocar a IDE correta!)

Configuração DISCOVERY (F407)

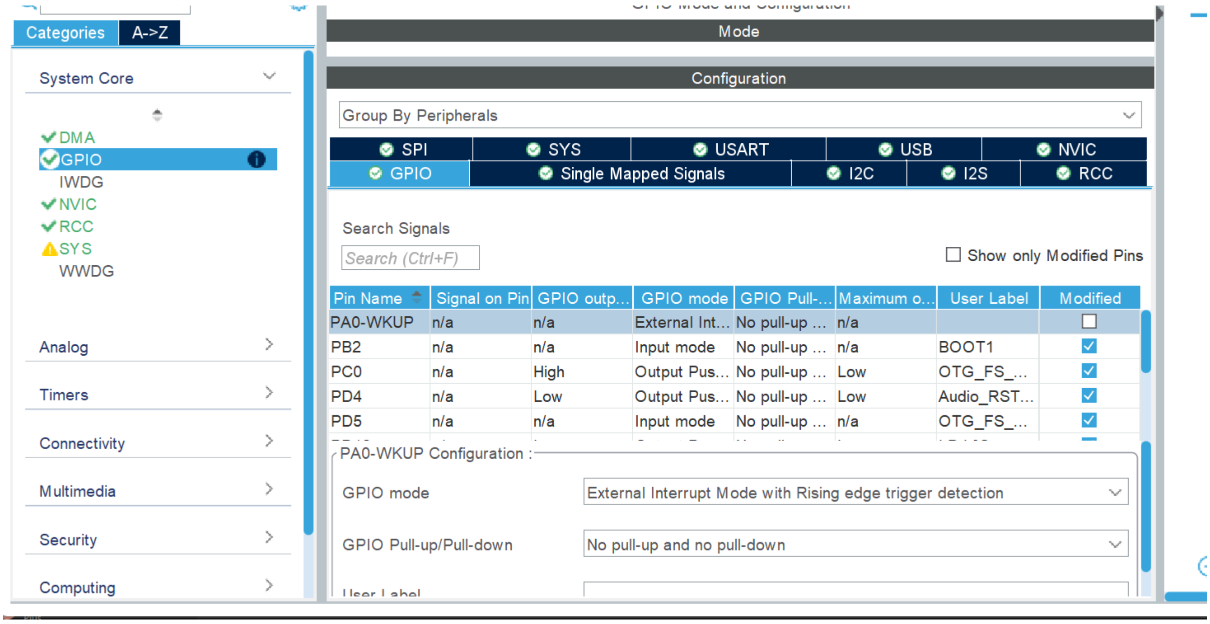


Coloquei o botão para ser o EXTi

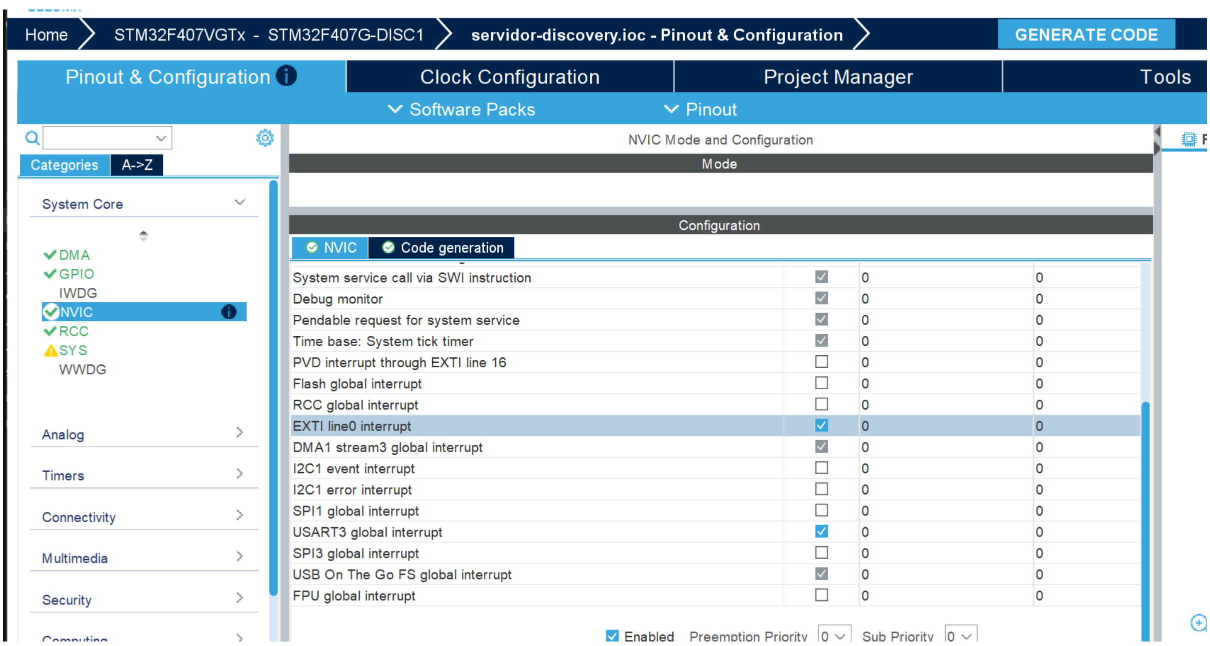




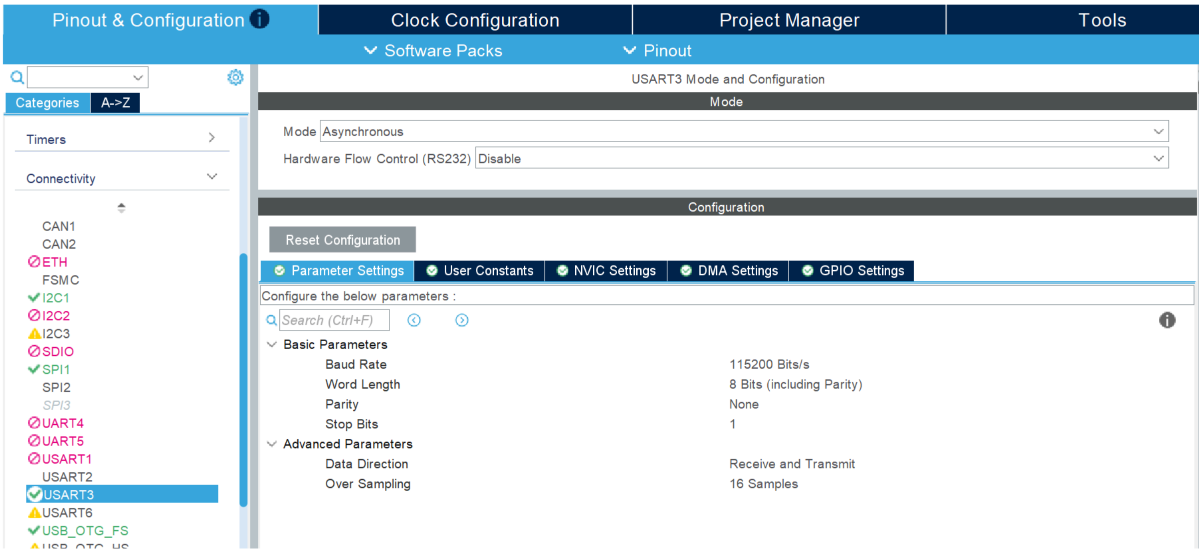
conferimos como o GPIO está:

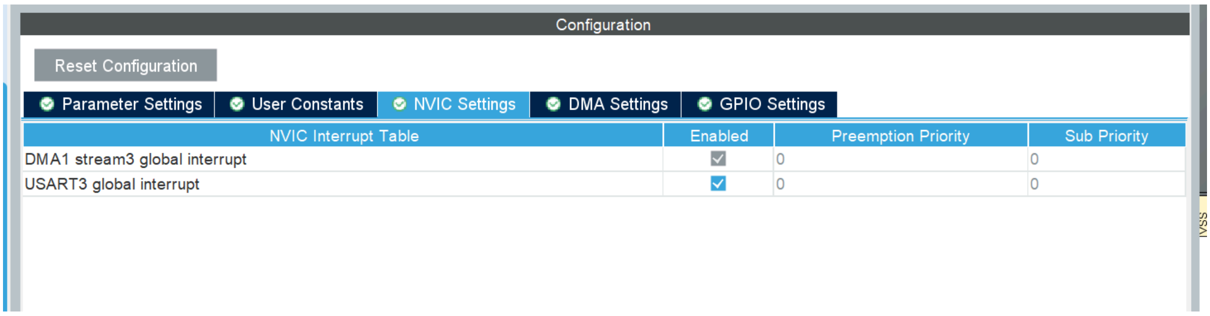


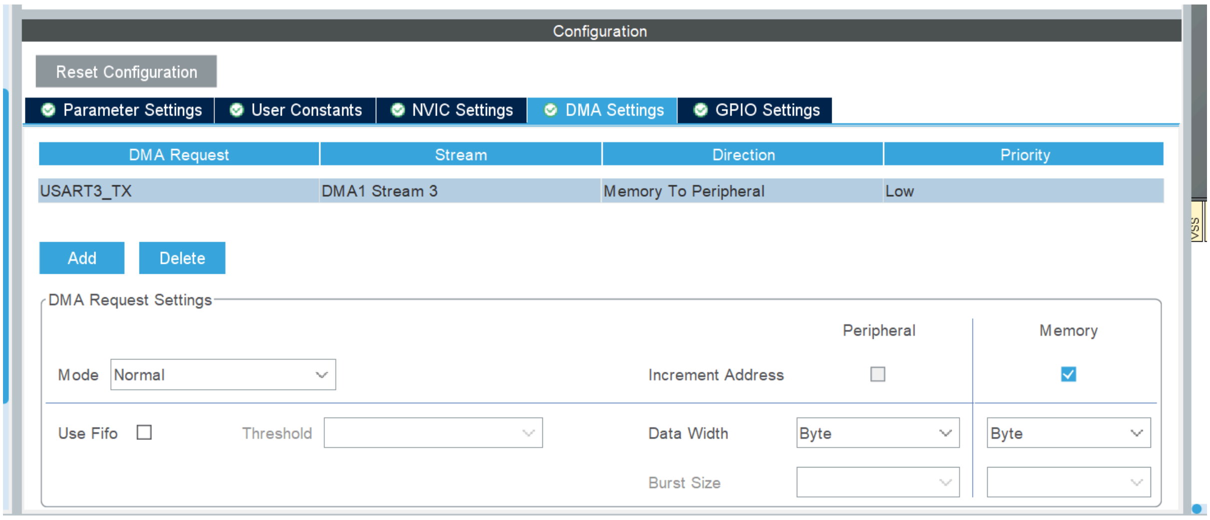
e marcamos a opção do EXTi

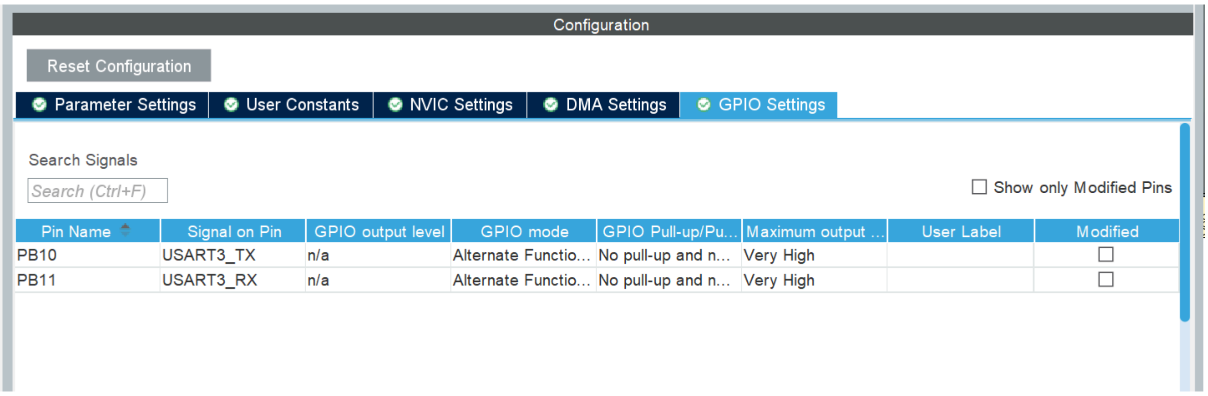


COMUNICAÇÃO:









Ligações físicas

Dados:

Núcleo GPIO D2(PA10) [USART1_RX] — jumper roxo----> [USART3_TX] PB10 Discovery.
Núcleo GPIO D8(PA9) [USART1_TX] — jumper cinza---> [USART3_RX] PB11 Discovery.

Alimentação:

Para gravar: ambas ligadas no USB

Para rodar o Projeto:

- 3v3 da nucleo —jumper marrom—> 3v da Discovery
- GND da Nucleo —jumper preto—> GND da discovery

(seguimos nesse modelo como solicitação o prof. em sala.).

Codigo

Codigo Nucleo

```c
/* USER CODE BEGIN 4 */
void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    if(GPIO_Pin == GPIO_PIN_13)
    {
        sprintf(log_buffer, "[BTN] Botao pressionado - enviando 0x5A...\r\n");
        log_msg(log_buffer);

        if(estado == IDLE)
        {
            HAL_UART_Transmit_IT(&huart1, &cmd, 1);

            sprintf(log_buffer, "[UART1 TX] Comando enviado.\r\n");
            log_msg(log_buffer);

            estado = WAITING_SERVER;
        }
        else
```

EXTi do botão

```c
        log_msg(log_buffer);
    }
    */

    // reativa recepção do contador
    HAL_UART_Receive_IT(&huart1, &contador_rx, 1);
}

void HAL_UART_RxCpltCallback_DMA(UART_HandleTypeDef *huart)
{
    if(huart->Instance == USART1)
    {
        sprintf(log_buffer, "[DMA RX] Tabela recebida!\r\n");
        log_msg(log_buffer);

        sprintf(log_buffer, "[TABELA RECEBIDA]:\r\n%s\r\n", tabela_rx);
```

reciver do contador na nucleo e Callback do DMA

```c
/* USER CODE BEGIN 3 */
  if(estado == PROCESSING)
  {
      sprintf(log_buffer, "[STATE] -> PROCESSING\r\n");
      log_msg(log_buffer);

      for(int i = 0; i < contador_rx; i++)
      {
          HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
          HAL_Delay(500);
      }

      // garante que termina apagado
      HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_RESET);

      sprintf(log_buffer, "[LED] Piscar concluido.\r\n");
      log_msg(log_buffer);
```

blink do led, feito por toggle (daí as piscadas pela metade. é um acendendo e outro apagando)

Codigo Discovery

```c
}

/* USER CODE BEGIN 4 */
void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    if(GPIO_Pin == GPIO_PIN_0)
    {
        contador++;

        // só toggle, sem delay
        HAL_GPIO_TogglePin(GPIOD, GPIO_PIN_12);
    }
}

void HAL_UART_RxCpltCallback(UART_HandleTypeDef *huart)
{
    if(huart->Instance == USART3)
    {
        if(rx == 0x5A)
        {
```

codigo e toggle do led.

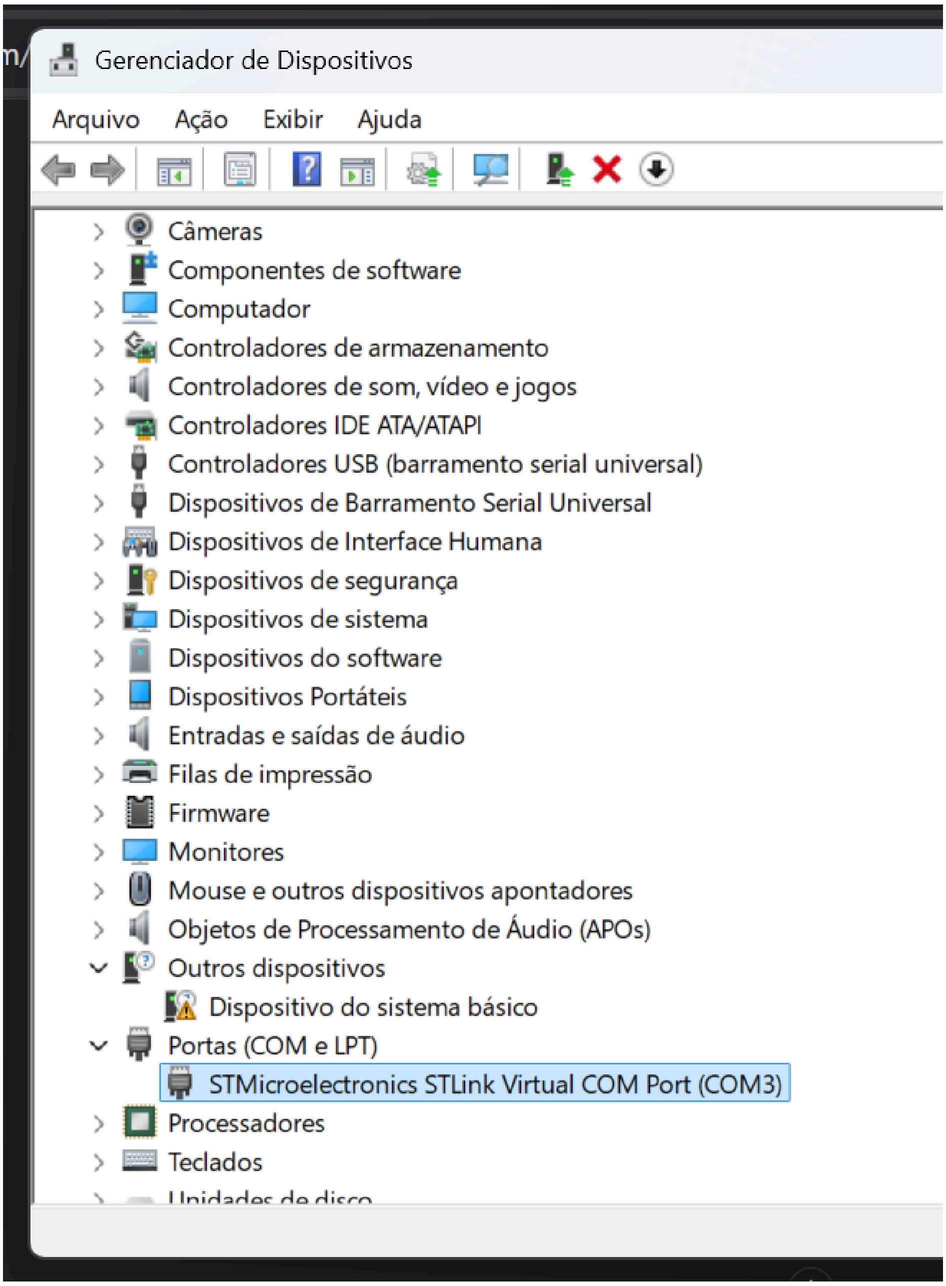
```c
    }
}

void HAL_UART_RxCpltCallback(UART_HandleTypeDef *huart)
{
    if(huart->Instance == USART3)
    {
        if(rx == 0x5A)
        {
            //HAL_UART_Transmit_IT(&huart3, &contador, 1);
            //HAL_UART_Transmit_DMA(&huart3, (uint8_t*)tabela, sizeof(tabela));
            HAL_UART_Transmit(&huart3, &contador, 1, 100);
            HAL_Delay(10); // pequeno delay

            //HAL_UART_Transmit(&huart3, (uint8_t*)tabela, sizeof(tabela), 100);
            uint16_t tamanho = strlen(tabela);
            HAL_UART_Transmit(&huart3, (uint8_t*)&tamanho, 2, 100);
            HAL_UART_Transmit(&huart3, (uint8_t*)tabela, tamanho, 100);

```

recebe a senha e faz o envio do contador e tabela.

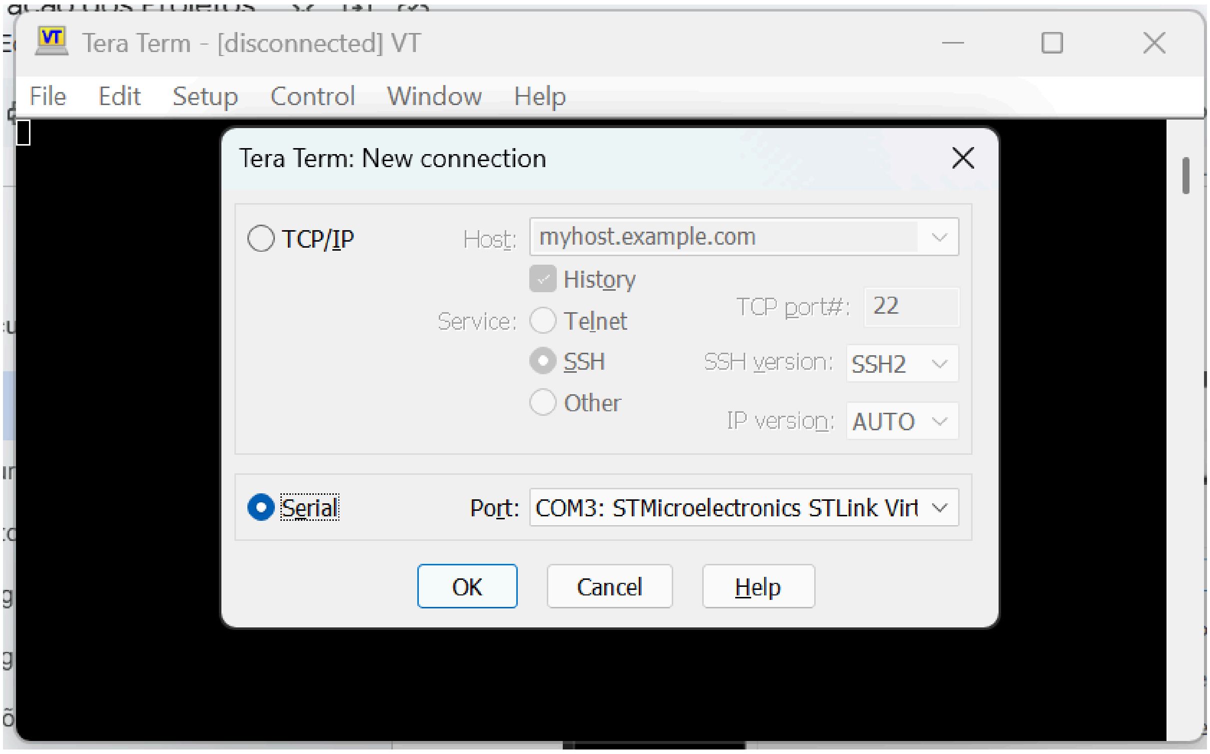
Teste

drivers: https://www.st.com/en/development-tools/stsw-link009.html
Advanced serial port terminal: https://www.com-port-monitoring.com/serial-port-terminal/

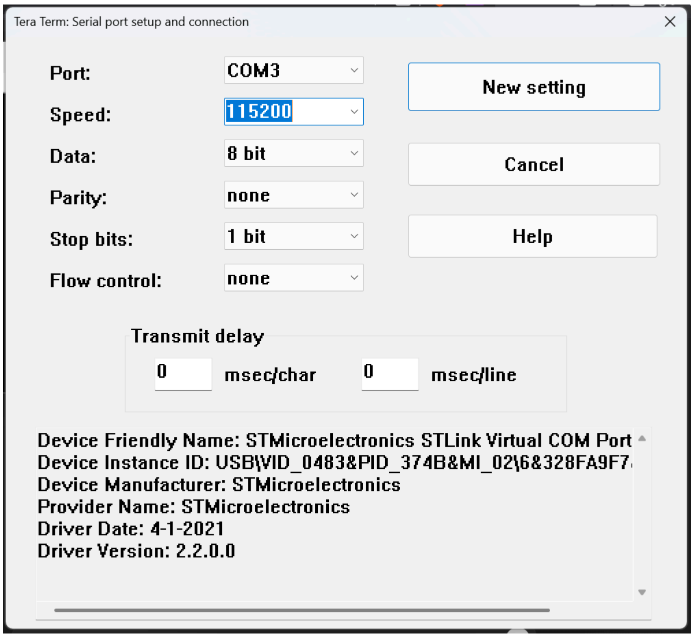
1. Localize a porta COM



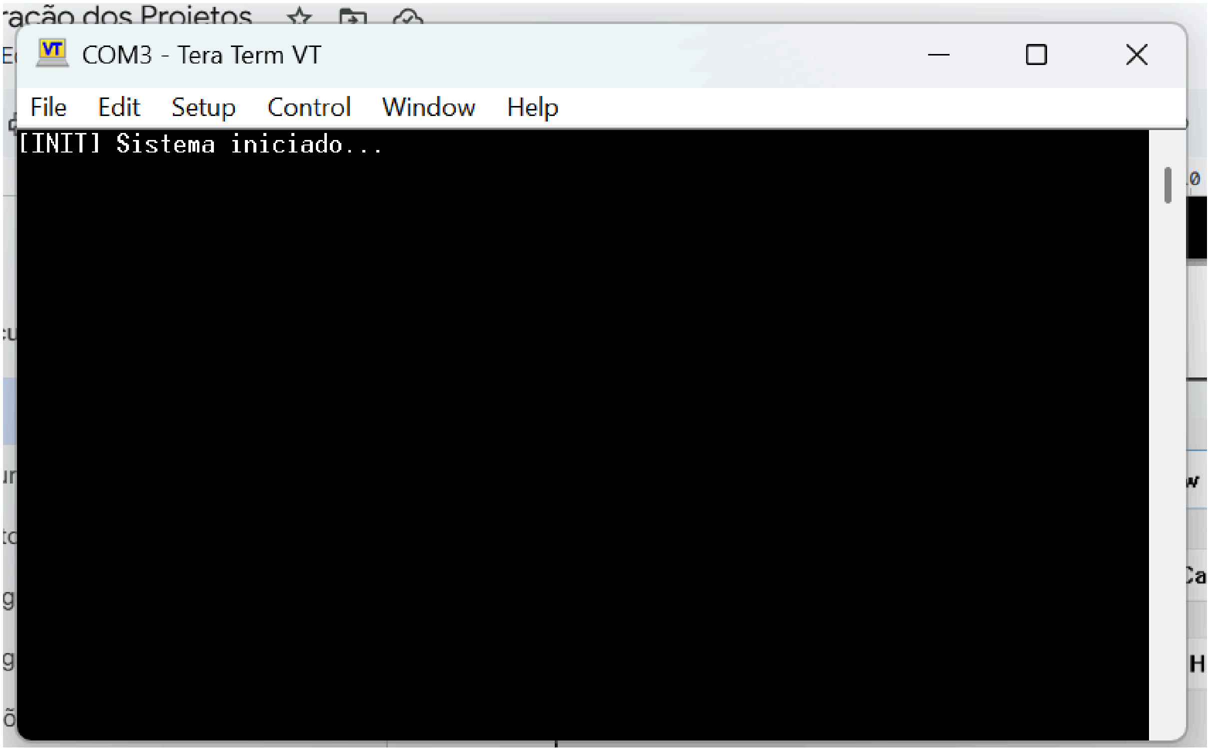
2. Preencha a new session



3. bauld rate // speed



4. codigo de inicialização



5. qnt de botões apertadas recebido